

DES 300 Digital Engineering Training

Training Report

Thai Airways International Public Company Limited

Submitted to

Digital Engineering Program (DE)

Sirindhorn International Institute of Technology

Thammasat University

By

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ACKNOWLEDGEMENT

I would like to sincerely thank the Digital Engineering Program at SIIT for providing me with the opportunity to undertake this enriching internship. I am deeply grateful to my internship supervisor and all team members at Thai Airways Cargo for their consistent guidance, warm welcome, and the trust they placed in me throughout the training period. Their advice, support, and constructive feedback played a crucial role in helping me overcome challenges and enhance both my technical and interpersonal skills.

This internship experience has been incredibly valuable, offering me insight into the real-world application of the skills I have learned in the classroom. It has also taught me how to work in a professional environment, adapt to responsibilities, and communicate effectively within a team. I truly appreciate the opportunities I was given to contribute meaningfully to ongoing projects. This experience has not only deepened my knowledge but also inspired me to pursue future goals with greater confidence and clarity.

EXECUTIVE SUMMARY

During my internship at Thai Airways Cargo, I had the chance to participate in a range of meaningful tasks that integrated both creative and technical responsibilities. My work focused on four main projects that were both challenging and rewarding. I collaborated with teammates on each assignment, such as sharing product photography duties for the catalogs and dividing the hurdle rate analysis by flight number—where I was responsible for TG03, TG04, and TG05.

The first project involved designing two separate product catalogs using Canva—one aimed at internal staff with more affordable and varied items, and another tailored for external clients with a refined and premium selection. I worked on this project in collaboration with colleagues in the department, sharing ideas and dividing responsibilities to ensure a well-structured and visually appealing outcome. I personally contributed by photographing the products, editing images, and helping to lay out the final design.

The second task was conducting product research to identify new merchandise suitable for sale, either to customers or for internal corporate use. I reached out to potential suppliers, gathered pricing information, and compiled all relevant details—including contact information and delivery conditions—into a comprehensive proposal document.

The third and most technically demanding assignment was developing a Python program to automate the analysis and reporting of cargo hurdle rates. This involved working with large datasets, merging fuel surcharge and airport data, calculating per-kilogram rates, and exporting everything into a formatted Excel report using openpyxl. The final output helped streamline what would otherwise be a time-consuming manual task.

The fourth project was developing a custom inventory management system for internal use. The website consists of three main pages: a Product Catalog page (which I developed), a Sales Log page, and a Report page. The catalog allows staff to add, edit, or delete item data such as product name, size, color, price, and stock levels. The system

supports daily and monthly sales reporting and allows the export of sales data as Excel files. I collaborated with two teammates on the system, with each of us responsible for one specific interface.

Together, these tasks enabled me to grow in multiple dimensions: creativity, business communication, data automation, and time management. They reflect the diverse skill set I've been developing and demonstrate my ability to contribute meaningfully in a professional environment.

CHAPTER 1

INTRODUCTION OF COMPANY

1.1 Company Profile

Thai Cargo is Thailand's air freight service provider, offering cargo transport to various destinations worldwide. Headquartered in Bangkok, the company stands out for its modern warehouses and secure operational processes. Thai Cargo focuses on delivering fast and reliable logistics services, supported by a high-quality service standard and an extensive flight network. The company operates a large fleet capable of handling a wide range of cargo types, including heavy-duty freighters and specialized shipping options such as express delivery, temperature-controlled transport, and high-value shipments. In addition, Thai Cargo features a modern tracking system and advanced IT infrastructure that allows customers to securely monitor the status of their shipments in real time.

Thai Airways Cargo is managed under Thai Airways International and operates with a centralized logistics control center. Its production capacity includes multiple aircraft types and dedicated cargo space for diverse shipment categories, ranging from perishables to oversized freight. The company's ground services, warehouse management systems, and automated tracking infrastructure contribute to its efficient handling and high throughput volume on a daily basis.

1.2 Company Contact Address and Map

Address Thai Airways International Public Company Limited

6th floor, Operation Center Building (OPC), Suvarnabhumi Airport 333 Moo 1, Nong Prue,

Bangphli, Samut Prakarn 10540

Tel (662) 137 4200 (24 hrs)

E-mail cargo.info@thaiairways.com

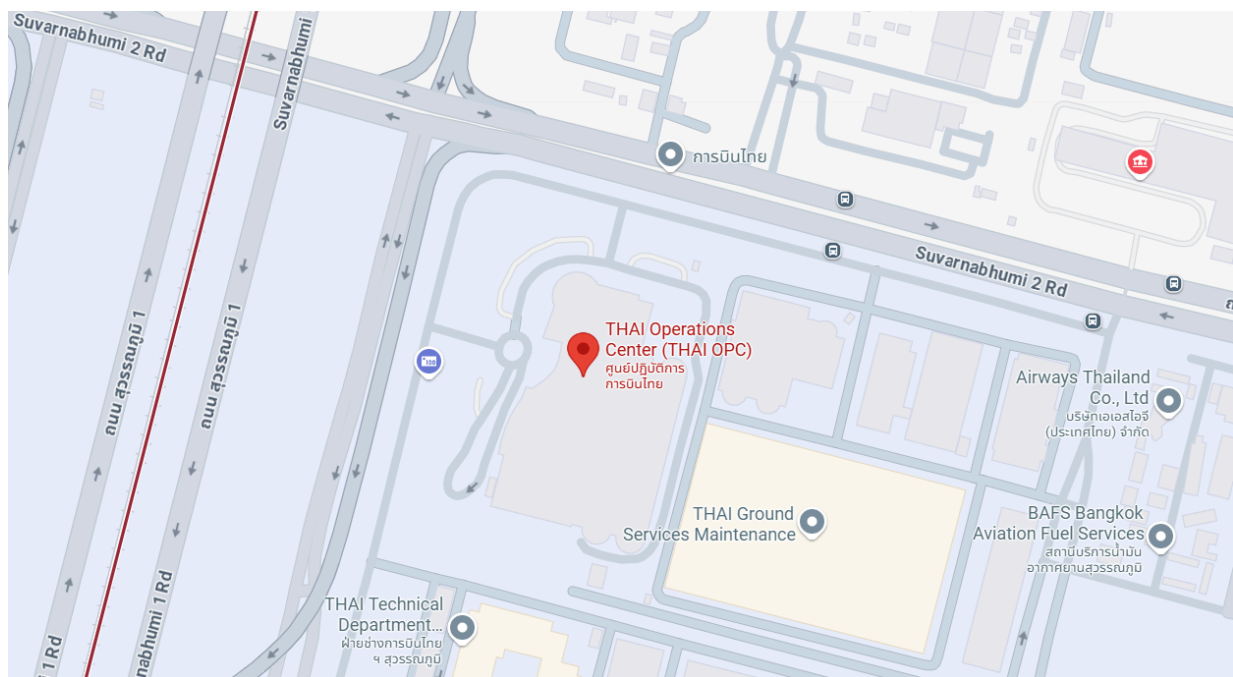


Figure 1: Map showing the location of Thai Airways Cargo Operations Center (OPC), Suvarnabhumi Airport.

1.3 Training Supervisor

Dhasimon Wimuktalaba (Market Research Analysis)

Tel 0824880884

E-mail dhaisamon.w@thaiairways.com

CHAPTER 2

TRAINING OBJECTIVES AND SCHEDULE

2.1 Training Objectives

- To gain hands-on experience with real business operations
- To develop practical skills in product design, research, and data automation
- To understand how logistics operations are managed within the aviation industry

2.2 Detail Description of Assigned Tasks during the Training Period

1. Designing Product Catalogs

I created two catalogs using Canva:

- **Giveaways Catalog:** Aimed at internal staff with more items and lower prices
- **Thai Airways Cargo Catalog:** Created for external customers with premium pricing I collaborated with two other interns in the department. Together, we divided the product photography responsibilities and supported each other throughout the project. My specific contributions included photographing items, editing images, and assisting with layout design to ensure everything aligned with the company's visual style.



Figure 2: Cover page of the Giveaways Catalog, designed using Canva

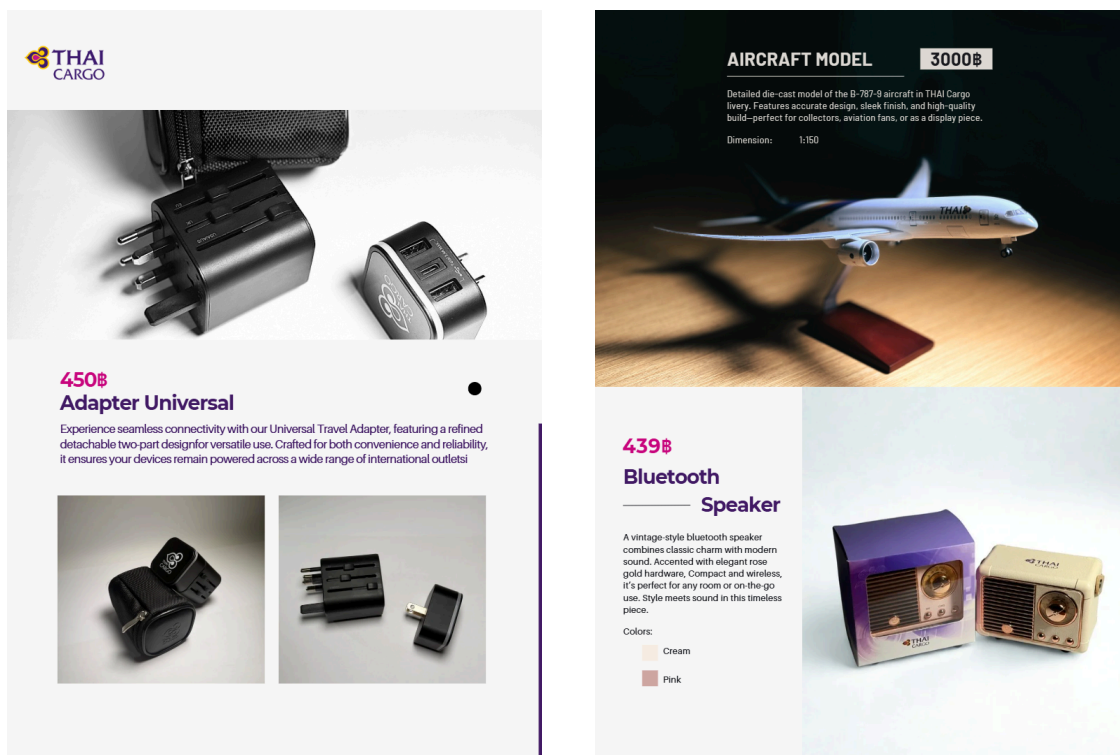


Figure 3: Sample pages from the product catalogs designed using Canva.

2. Product Proposal Research

I prepared a product proposal document titled "THAICargoProductsrecommend.pdf". To complete it, I reached out to suppliers, gathered quotes, collected contact details, and ensured that each item fit the company's branding and needs.

Product	Price	Source
 ผ้าอเนกประสงค์ (Microfiber)	แฉกรถบรรทุกซีดแก้ว - ผ้า poly 80/20 - ขนาด 15*18 cm - จำนวน 100 ผืน - แพคเกจ 100/1 ราคาชิ้นละ 36 บาท = 3600 บาท (100 ชิ้น) *ราคานี้ไม่รวมภาษีมูลค่าเพิ่ม *ราคารับรวมส่งพัสดุหิ้ว : 1 ครั้ง	บริษัท ภาพพิมพ์ 142 จำกัด www.phap-pim142.com Phone: 0841539194 Email: sale@phap-pim142.com Line : @phappim142
 นาฬิกา	100 ชิ้น 255 บาทต่อหน่วย = 25500 บาท (100 ชิ้น)	Grand Premium Marketing Company Limited www.grand-premium.com Phone: 0827007432-3 Email : info@grand-premium.com Line : @grandpremium
 แก้วน้ำพับได้	กระบอกน้ำพรีมียม ความจุ 420ml สกรีน 1 สี : ดำ เทม่ง จำนวนชิ้นต่อ 100 ชิ้น ราคา 17,999 บาท *ราคาไม่รวม VAT และ จัดส่ง ใช้ระยะเวลาในการผลิต 30-40 วันทำการ	บริษัท คิงคองพรีเมียมเทรดดิ้ง จำกัด www.kingkongpremium.com Phone: 0954946261 Email : Kingkongpremium@gmail.com Line : @Kingkongpremium
 กระเป๋าใส่เหรียญ (ผ้าขนสั้น)	มากกว่า 50 ชิ้น ชิ้นละ 90 บาท = 9000 บาท (100 ชิ้น)	iDuckyOfficial www.iduckyofficial.com Phone: 0965699414 Email : iduckyshop03@gmail.com Line : @iduckyofficial
 ไฟฉาย N1035	100 ชิ้น 125 บาทต่อหน่วย = 12500 บาท (100 ชิ้น)	บริษัท เกรทพรีเมียม จำกัด www.greatpremium.co.th Phone: 0219226468 Email : Premium88@gmail.com Line : @greatpremium

Figure 4: Product proposal summary page created for Thai Airways Cargo, including item descriptions, prices, and supplier contact details.

3. Python Tool for Hurdle Rate Analysis

This project was a collaborative effort between myself and two other interns. We divided the workload based on flight number ranges. I was responsible for handling data related to flight numbers TG03, TG04, and TG05, while my teammates worked on TG06 and TG09. I wrote a Python script called CompletedCombineHurdleRate.py that:

- Processes flight data from large CSV files
- Merges fuel charge and airport data
- Calculates rates and surcharges for each shipment
- Outputs a styled Excel file (Completed Combine Hurdle Rates.xlsx) summarizing all flights This tool saved significant time and eliminated the need for manual calculations.

```
C:\> MENT > TGwork > Code > HurdleRateCalculation > CompletedCombineHurdleRate.py > ...
1 import pandas as pd
2 from openpyxl import Workbook
3 from openpyxl.styles import PatternFill, Alignment, Font, Border, Side
4 from openpyxl.utils import get_column_letter
5
6 # --- FILE PATHS ---
7 # Ensure these files are in the same directory as your script
8 # or update the paths to their correct locations.
9 tg_data_path = r"C:\MENT\TGwork\Updated(with flight no.)\Data Inputs WorldACD 2025 JAN-JUN_30_904951609924539164.csv"
10 fuel_list_path = r"C:\MENT\TGwork\Code\DataFiles\Fuel\SeperatedCSV\Fuel Lists.csv"
11 airport_list_path = r"C:\MENT\TGwork\Code\DataFiles\Fuel\SeperatedCSV\Airport List.csv"
12 output_excel_path = r"C:\MENT\TGwork\Code\Complete\Completed Combine Hurdle Rates.xlsx" # Updated output name for clarity
13
14 # --- DEBUGGING OUTPUT PATHS ---
15 debug_output_dir = r"C:\MENT\TGwork\Code\Complete\extra\table" # Directory to save debug CSVs
16 debug_fuel_expanded_path = debug_output_dir + "df_fuel_expanded_debug.csv"
17 debug_fuel_processed_path = debug_output_dir + "df_fuel_processed_debug.csv"
18 debug_df_after_fuel_merge_path = debug_output_dir + "df_after_fuel_merge_debug.csv"
19
20
21
22 # --- LOAD MAIN DATA ---
23 try:
24     df = pd.read_csv(tg_data_path)
25     print(f"Successfully loaded {tg_data_path}. Initial DataFrame head:\n{df.head()}")
26 except FileNotFoundError:
27     print(f"Error: The file '{tg_data_path}' was not found. Please ensure it's in the correct directory.")
28     exit()
29
30 # --- LOAD FUEL AND AIRPORT LISTS ---
31
32 # --- PREPROCESSING FOR MAIN DATA (df) ---
33 df['awb_nr'] = df['awb_nr'].astype(str)
34 df['Flight Number'] = df['Flight Number'].astype(str)
35 # Convert and standardize airport codes for merging
36 df['Board Point Airport Code'] = df['Board Point Airport Code'].astype(str).str.strip().str.upper()
37 df['Off Point Airport Code'] = df['Off Point Airport Code'].astype(str).str.strip().str.upper()
38
39 # Convert numeric columns, handling errors by coercing to NaN
40 df['net_freight charges'] = pd.to_numeric(df['net_freight charges'], errors='coerce')
41 df['chargeable_weight'] = pd.to_numeric(df['chargeable_weight'], errors='coerce')
42
43 # Filter out rows where net_freight charges are not positive
44 df = df[df['net_freight charges'] > 0].copy()
45
46 # Calculate rate_per_kg and rate_usd for individual legs initially
47 df['rate_per_kg'] = df['net_freight charges'] / df['chargeable_weight']
48 exchange_rate = 0.031
49 df['rate_usd'] = df['rate_per_kg'] * exchange_rate
```

Figure 5: Example Python Code from the Hurdle Rate Calculation Script

Flight: TG0301									
WEIGHT BREAK	N	45	100	500	1000	2000	AvgFreight	All-In	AvgFuel
WT.RATIO	28%	20%	36%	12%	2%	2%	100%		
MAX	81.82	9.25	35.42	5.47	3.58	4	35.42	35.75	
MEAN	8.5	2.43	1.71	1.23	1.34	1.06	1.8	2.13	0.33
MIN	1.44	0.62	0.79	0.56	0.74	0.4	0.4	0.73	
the Lowest rate	DELGRN	BKKGRN	ICNRGN	BKKGRN	BKKGRN	ICNRGN	ICNRGN		
Origin + Dest	N	45	100	500	1000	2000	AvgFreight	All-In	Fuel
AMDRGN	-	-	1.58	-	-	0.72	1.15	1.54	0.40
ARNRGN	-	-	1.18	-	-	-	1.18	1.75	0.57
BKKRGN	8.39	1.85	1.22	1.05	0.94	1.13	1.18	1.58	0.40
BLRRGN	-	-	-	-	-	0.64	0.64	1.03	0.40
BOMRGN	-	1.38	1.49	-	-	1.23	1.42	1.81	0.40
BRURGN	3.89	2.35	35.42	-	-	-	18.89	18.89	Gross
BSLRGN	11.86	-	-	-	-	-	11.86	12.41	0.55
CCURGN	-	-	1.11	-	-	-	1.11	1.5	0.40
CDGRGN	2.46	2.35	-	-	-	-	2.35	2.97	0.62
CGKRGN	6.01	3.15	1.27	0.95	0.92	0.92	2.42	2.61	0.19
CNKRGN	-	-	-	-	1.43	-	1.43	1.83	0.40
DACRGN	3.68	2.03	1.51	-	-	1.15	1.65	2.15	0.50
DELGRN	1.46	1.35	1.24	0.97	0.97	1.11	1.14	1.53	0.40
DXBRGN	-	3.45	-	-	-	-	3.45	4.05	0.60
FRARGN	25.26	6.04	3.4	-	-	-	4.84	5.53	0.69
GVARGN	4.39	-	4.12	-	-	-	4.12	4.67	0.55
HKGRGN	-	1.47	1.35	1.32	-	0.86	1.41	1.49	0.08
HNDGRN	11.27	4.95	4.88	-	-	-	4.9	5.16	0.26
HYDRGN	-	-	1.54	1.41	-	1.01	1.41	1.81	0.40
ICNRGN	5.19	1.2	0.8	0.8	0.81	0.47	0.88	1.06	0.18
ISBRGN	-	-	-	1.37	1.37	1.21	1.27	1.56	0.29
KHHRGN	-	-	1.66	-	-	-	1.66	1.66	Inclusive
KHIRGN	-	1.48	1.74	1.4	-	-	1.48	1.77	0.29
KXRGN	21.6	3.55	3.62	3.54	2.74	-	3.55	3.81	0.26
KULRGN	2.8	-	1.19	-	-	-	1.19	1.46	0.27
LHERGN	-	-	1.6	-	-	-	1.6	1.89	0.29
LHRRGN	14.35	-	-	-	-	-	14.35	15.08	0.73
MAARGN	1.57	-	-	-	0.92	-	0.92	1.32	0.40
MILRGN	3.88	3.24	2.18	-	-	-	2.71	2.71	Inclusive
MUCRGN	-	-	-	-	3.58	-	3.58	4.27	0.69
NGORGN	12.12	3.07	2.7	2.82	2.91	-	2.88	3.14	0.26
NRTRGN	10.47	4.72	4.8	5.47	-	-	4.77	5.03	0.26
PEKRGN	-	1.28	1.05	-	-	-	1.24	1.4	0.16
PERRGN	-	2.53	0.91	-	-	-	2.21	2.55	0.34
PVGRGN	-	2.31	1.89	1.2	1.32	-	1.92	2.08	0.16
SGHRGN	-	-	1.68	-	-	-	1.68	2.05	0.37

Figure 6: Example Output Table from the Hurdle Rate Report

4. Inventory Management System Development

I also developed an inventory management system tailored for internal organizational use. My main responsibility was building the interface for Product Catalog Page for adding and editing product items in the catalog. The interface allows staff to input details such as product name, size, color, stock levels, and pricing. I implemented these features using Streamlit to ensure a simple and user-friendly experience.

The full system enables users to:

- Add, delete, and edit items in the inventory (my specific contribution)
- Record and remove sales logs
- View sales reports on a daily and monthly basis
- Export sales data as downloadable PDF files

This system was designed to help streamline internal product tracking and improve the overall efficiency of organizational sales operations.



Manage all product inventory with sizes, colors, and pricing options.

 All Products

	Product ID	Product Name	Size	Color	Stock	Gifts&More Shop Price	Requisition Price	Stock Date	Status	Remark
0	NA001	Adapter Universal	Default	Default	99	570	450	02 JUL 2025	Active	-
1	NA002	Aircraft Model Premium (B747-400 freighter)	Default	Default	100	-	2140	02 JUL 2025	Active	-
2	NA003	Aircraft Model (B787-900)	Default	Default	100	4000	3000	07 JUL 2025	Active	-

Action

☒ Add Product ☐ Edit Product ☐ Delete Product

+ Add New Product

Product Name

Initial Stock

Figure 7: Example UI of the Inventory Management System Web Interface (Product Catalog Page)

2.3 Training Schedule

Daily Schedule

Weeks of Training	Assigned Task	Description
Week1		
09/06/2025	- Orientation day	- Attended orientation at the head office
10/06/2025	- Took a picture of each product	- Took photos of each product for the Thai Cargo catalog
11/06/2025	- Prepared internship documents and assisted with the model	- Worked on the end-to-end cargo operations and revenue model for Thai Future Logistics
12/06/2025	- Continued taking photos of each product	- Took photos of each product for the Thai Cargo catalog
13/06/2025	- Attended a conference with my boss	- Attended a meeting with my boss and colleagues to receive assigned tasks
Week2		
16/06/2025	- Researched terms assigned by the supervisor	- Looked up the meaning of terms such as “E-commerce of the Earth Market,” “E-commerce of the Asia Market,” and “BKK Market”
17/06/2025	- Continued taking photos of each product	- Took photos of each product to create the Thai Cargo catalog
18/06/2025	- Began the task assigned during the conference	- Started working on the hurdle rate calculation program

19/06/2025	- Continued developing the hurdle rate calculation	- Sorted each flight and boarding point to simplify processing - Calculated the rate per kilogram for each flight
20/06/2025	- Continued developing the hurdle rate calculation	- Wrote Python code for the hurdle rate calculation program
Week3		
23/06/2025	- Continued taking photos of each product	- Took photos of each product for use in the Thai Cargo catalog
24/06/2025	- Continued taking photos of each product	- Took photos of each product for use in the Thai Cargo catalog
25/06/2025	- Continued taking photos of each product	- Took photos of each product for use in the Thai Cargo catalog
26/06/2025	- Continued taking photos of each product	- Took photos of each product and began creating the catalog using Canva
27/06/2025	- Started creating the catalog using Canva	- Inserted the previously taken product images into the catalog in Canva

Week 4		
30/06/2025	- Continued creating the product catalog using Canva	- Inserted the previously taken product images into the catalog and wrote product descriptions
1/07/2025	- Continued creating the product catalog using Canva	- Inserted the previously taken product images into the catalog and wrote product descriptions
02/07/2025	- Researched products to recommend to the boss	- Researched products to propose to Thai Cargo for potential sale in the company store
03/07/2025	- Attended conference with boss and began inventory program	- Attended a conference where a company presented an inventory system - Used code to begin developing the inventory program for cargo operations
04/07/2025	- Go to the conference room with the boss	- Go to the conference room with the boss to help them to edit the presentation slide that they need to present to others.
Week 5		
07/07/2025	- Attended a meeting at the conference room with the boss	- Assisted the boss in editing presentation slides for an upcoming presentation
08/07/2025	- Visited the head office in Bangkok with the boss	- Helped the boss edit presentation slides at the head office conference room

9/07/2025	- Attended a meeting at the conference room with the boss	- Assisted the boss in editing presentation slides for an upcoming presentation
Week 6		
14/07/2025	- Created user guide documents for the inventory system program	- Wrote user documentation to assist users in operating the inventory system
15/07/2025	- Continued developing the hurdle rate calculation	- Wrote Python code for the hurdle rate calculation program
16/07/2025	- Continued developing the hurdle rate calculation	- Wrote Python code for the hurdle rate calculation program
17/07/2025	- Leave of absence due to an accident	- Leave of absence due to an accident
18/07/2025	- Continued developing the hurdle rate calculation	- Wrote Python code for the hurdle rate calculation program
Week 7		
21/07/2025	- Continued developing the hurdle rate calculation	- Wrote Python code for the hurdle rate calculation program
22/07/2025	- Continued developing the hurdle rate calculation	- Wrote Python code for the hurdle rate calculation program
23/07/2025	- Continued developing the hurdle rate calculation	- Wrote Python code for the hurdle rate calculation program
24/07/2025	- Continued developing the hurdle rate calculation	- Wrote Python code for the hurdle rate calculation program

25/07/2025	- Continued developing the hurdle rate calculation	- Wrote Python code for the hurdle rate calculation program
Week 8		
29/07/2025	- Presented all completed work to a professor from Thammasat University	- Gave a summary presentation of all tasks completed during the internship
30/07/2022	- Presented all completed work to the department supervisor and team	- Gave a final presentation of all work done throughout the internship to the department

CHAPTER 3

ACHIEVEMENT

3.1 Overall Achievement

I successfully completed all assigned projects, each of which presented its own unique challenges and learning opportunities. From creative tasks like product photography and catalog design to technical programming and data automation, I was able to apply both design and engineering skills. Working collaboratively on multiple projects also enhanced my ability to communicate, plan, and contribute within a team setting. Most importantly, I gained hands-on experience with real business processes, developed solutions that were actually used by the team, and saw how my work directly impacted internal efficiency and decision-making.



Figure 8: Photograph of the product photography session conducted for the catalog project during the internship.

3.2 Technical Achievement

Throughout my internship, I developed and refined a wide range of technical skills. I became proficient in using Canva to design polished product catalogs that aligned with the company's branding standards. For the Hurdle Rate project, I wrote a Python script using the pandas and openpyxl libraries to process large datasets. The program cleaned, analyzed, and exported data into Excel with formatted layouts, color styling, and merged cells to match the organization's reporting format. I also implemented logic to map fuel surcharges according to geographic regions, airport codes, and flight numbers to ensure accurate and automated summaries.

Additionally, I developed the Product Catalog page of a custom inventory management system using Streamlit. This page allows staff to add, edit, and delete product records such as item name, size, color, price, and stock quantity through a user-friendly interface. I was fully responsible for designing and implementing this page, from front-end layout to back-end logic and data handling. While the full system included other modules like Sales Logging and Reporting (developed by my teammates), my work on the catalog page played a crucial role in enabling smooth inventory control and data consistency across the platform.

ไฟล์

หน้าแรก

แทรก

เค้าโครงหน้ากระดาษ

สูตร

ข้อมูล

รีวิว

มุมมอง

ACROBAT

Terabox

ค้นหา

ลบ

คัดลอก

วาง

การตั้งค่าการรูปแบบ

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Figure 9: Excel output generated by the Python program for hurdle rate analysis. The report includes calculated fuel surcharges and formatted layouts based on flight data.

3.3 Soft Skills Achievement

During my internship, I developed several essential soft skills that contributed to effective collaboration and productivity in a professional setting. I worked closely with peers to co-develop projects like the product catalogs and hurdle rate analysis, which enhanced my teamwork and adaptability. I strengthened my problem-solving abilities by addressing bugs, making design decisions, and responding to challenges as they arose. Clear communication and proper documentation were key in ensuring smooth coordination within the team. Additionally, I improved my time management skills by setting priorities, meeting deadlines, and adjusting to changing project needs efficiently.

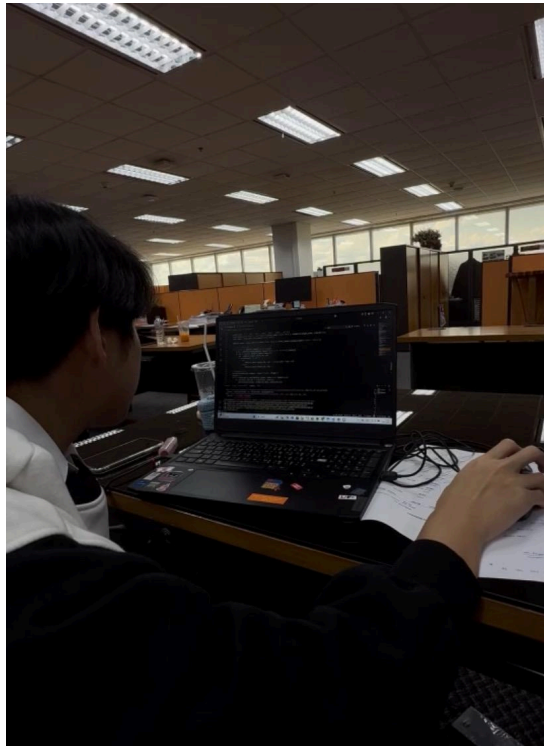


Figure 10: Working with teammates to debug and test features during the development of the inventory management system, reflecting collaborative problem-solving and communication.

CHAPTER 4

CONCLUSION AND RECOMMENDATIONS

To sum up, my internship at Thai Airways Cargo was an incredibly valuable experience that contributed significantly to both my technical and personal development. I had the opportunity to take part in four core projects: designing product catalogs, creating a supplier product proposal, developing a hurdle rate automation tool in Python, and building an internal inventory management system. Among these, I played a key role in developing the Product Catalog page, which served as a central interface for inventory control and data input.

Each project presented its own challenges and learning opportunities, allowing me to enhance a variety of technical skills. I became proficient in tools such as Canva, Python (with pandas and openpyxl), and Streamlit for developing functional web-based systems. I also gained hands-on experience working with real-world datasets, automating manual processes, and designing user interfaces with a focus on simplicity and usability. Beyond the technical aspects, I learned how to collaborate effectively, divide tasks among team members, and adapt to evolving project needs in a professional setting.

As a recommendation for future interns, I would encourage them to stay curious and proactive—ask questions when uncertain, document their progress clearly, and seek opportunities to contribute beyond the assigned responsibilities. Embracing feedback and staying open to new challenges are also essential for maximizing personal growth and delivering impactful work. By doing so, interns will not only accelerate their learning but also create meaningful value for the teams and organizations they work with.

Finally, I am truly grateful for the opportunity to intern at Thai Airways Cargo. This experience has deepened my understanding of how digital solutions can improve real-world operations and has inspired me to continue exploring the intersection between technology and business.

APPENDIX

1. Designing Product Catalogs

- **Description:** Created two catalogs using Canva:
 - *Giveaways Catalog*: for internal staff use with a wider product selection and affordable pricing
 - *Thai Airways Cargo Catalog*: for external clients, featuring selected premium products
- **Responsibilities:**
 - Took product photos with mobile phone cameras under indoor lighting
 - Edited and retouched photos
 - Helped design layout and product arrangement in Canva
- **Tools Used:**
 - **Canva** (design)
 - **Microsoft Word & PDF** (final formatting/export)
 - **Smartphone camera** (product photography)

2. Product Proposal Research

- **Description:** Created a document proposing merchandise for sale
- **Responsibilities:**
 - Researched suitable products
 - Contacted suppliers via phone and email
 - Collected prices, MOQ, delivery terms, and contact info
 - Compiled all data into a proposal file (*THAICargoProductsrecommend.pdf*)

- **Tools Used:**
- **Google Sheets** (data collection)
- **Microsoft Word/PDF** (final document)
- **Email / Phone / Line** (supplier communication)

3. Hurdle Rate Calculation Tool

- **Description:** Wrote a Python script to calculate hurdle rates from raw flight data
- **Responsibilities:**
 - Handled flight numbers TG03, TG04, TG05
 - Parsed and merged flight, surcharge, and airport data
 - Automated calculations and exported results to Excel
 - Designed formatting, cell merging, and styling in final report
- **Tools Used:**
 - **Python 3**
 - **pandas** (data processing)
 - **openpyxl** (Excel export and styling)
 - **Excel** (for review and manual validation)

4. Inventory Management System Development

- **Description:** Developed part of an internal web-based inventory system using Streamlit

- **Responsibilities:**

- Designed and implemented the **Product Catalog Page**
- Enabled adding, editing, and deleting product records
- Handled UI layout, state management, and data validation

- **Tools Used:**

- **Python**
- **Streamlit**
- **pandas**
- **Excel** (as the backend storage format)